Interference Patterns: A Resonant Space

WPI

An Interactive Qualifying Project submitted to the faculty of
WORCESTER POLYTECHNIC INSTITUTE
in partial fulfillment of the requirements for the Degree of Bachelor of Science

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Abstract

Mental health concerns among college students have significantly increased in recent years due to academic stress, social isolation, and financial security fatigue. Worcester Polytechnic Institute (WPI) has experienced growing concerns regarding student anxiety, depression, and emotional burnout, particularly following the COVID-19 pandemic. The goal of this project was to design and implement an interactive mechanical art display space on the WPI campus intended to improve mental wellness through creativity, collaboration, and engineering engagement. Through background research, analysis of interested parties, and investigation of existing mental health initiatives, this project explored how public interactive art may encourage social interaction while providing emotional support within the technically focused university environment. The project proposes a collaborative student-centered art display designed to strengthen campus connection and provide a preventative approach to student wellness.

Acknowledgements

We would like to thank the Worcester Polytechnic Institute community for supporting this project and their perspectives regarding student wellness and campus culture. We also extend our gratitude to our advisor, faculty, and peers for their guidance throughout the development of this project.

Finally, we would like to acknowledge the students whose challenges and experiences have brought greater awareness to the importance of mental health care and student support on campus. Their impact has contributed to ongoing conversations, expanded resources, and continued efforts to improve the overall wellness of future generations of students.

Executive Summary

1. Introduction

The purpose of this project was to design and develop an immersive interactive art installation that combines engineering, artistic expression, and collaborative participation to support student mental wellness and social connectedness at Worcester Polytechnic Institute (WPI). The project explored how engineering-centered artistic environments and sensory-responsive public installations could function as preventative wellness spaces that encourage creativity, emotional engagement, personal reflection and interpersonal interaction within the campus community.

2. Background

Mental health concerns among university students have become an increasingly significant issue across higher education institutions. Academic pressure, social isolation, financial concerns, and uncertainty regarding future employment have contributed to rising levels of stress, anxiety, emotional burnout, and loneliness among students. At Worcester Polytechnic Institute, these concerns became especially visible during the 2021 and 2022 academic years, when students reported increased emotional distress and the campus community experienced several tragic losses connected to mental health struggles.

Although WPI implemented initiatives such as counseling resources, mental health days, modifications to the grading system, and community-centered spaces, many students continued to experience feelings of social disconnection and emotional exhaustion. Existing support systems largely focused on crisis response rather than preventative methods that encourage community interaction and emotional engagement before students reach severe levels of distress.

Research surrounding participatory art, immersive installations, and collaborative creative environments suggests that artistic engagement may positively influence emotional wellness, reduce stress, and strengthen social connectedness. Engineering-centered artistic systems additionally align closely with WPI's interdisciplinary and technical campus culture.

3. Methodology

The project, titled I*nterference Patterns: A Resonant Space*, utilized background research, participatory brainstorming sessions, collaborative design practices, engineering development, and fabrication to preliminarily investigate the relationship between immersive artistic environments and emotional wellness.

Research was conducted regarding:

- Student mental health trends,
- Participatory art and immersive installations,
- Collaborative community environments,
- Sensory-responsive artistic systems

Participatory brainstorming sessions encouraged students to collaboratively explore concepts of emotional connection, isolation, sensory interaction, and shared experiences. Students contributed conceptual ideas involving projections, motion-responsive systems, ambient soundscapes, tactile interaction, and collaborative artistic participation.

The project additionally emphasized interdisciplinary collaboration between engineering and artistic design. Routine communication meetings and collaborative construction helped guide conceptual development while ensuring that engineering systems remained connected to the broader emotional and social goals of the installation.

The design process combined collaborative ideation with engineering fabrication and artistic prototyping. Participatory sessions served as the conceptual foundation of the project by encouraging students to collectively contribute installation concepts and emotional wellness ideas. The primary physical deliverable of the project was the successful fabrication of an interactive wall structure that extends and retracts in response to participant proximity intended to serve as the foundational framework for a larger immersive installation environment.

4. Findings

Several major findings emerged throughout the project. Students expressed a strong desire for low-pressure collaborative environments outside of traditional academic settings. Immersive and interactive artistic systems appeared significantly more engaging than passive artistic displays, particularly when incorporating sensory responsiveness and collaborative participation.

The project also demonstrated that engineering-centered artistic environments align strongly with WPI's technical and interdisciplinary culture while encouraging creativity and emotional engagement. Additionally, the participatory design process itself strengthened interpersonal interaction and community involvement throughout the project.

Although the complete immersive installation was not finalized within the project timeline, the successful construction of the wall prototype demonstrated the feasibility of developing a larger engineering-centered wellness installation on the WPI campus.

5. Recommendations and Continued Development

The project recommends that WPI continue exploring immersive engineering-centered artistic environments as preventative wellness initiatives. Future continuation efforts should focus on:

- Sensor testing and responsive system integration
- Participant interaction studies
- PANAS-based emotional wellness evaluation
- Immersive projection systems
- Ambient soundscapes
- Expanded sensory interaction environments

Additional IQPs and interdisciplinary continuation projects are strongly recommended in order to progressively expand the installation toward its intended fully immersive state.

Long-term recommendations include the development of a permanent interactive wellness space, increased interdisciplinary collaboration, and continued research examining the relationship between engineering-centered public art and emotional wellness within university communities.

Conclusion

Interference Patterns: A Resonant Space demonstrated how engineering and artistic collaboration can extend beyond traditional technical applications to support emotional wellness, foster creativity, and strengthen social connectedness within higher education environments. While the current project primarily established the conceptual framework and foundational arches wall prototype of the installation, it successfully created a scalable platform for future continuation. Ultimately, the project highlights the potential for immersive engineering-centered artistic environments to serve as collaborative and preventative wellness spaces within university communities.

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1. Introduction

Mental health concerns among university students have drastically increased over the past decade. Increased academic pressure (Barbayannis et al. 2022), future employment uncertainty (Chen et al., 2021), financial concerns (Moore et al., 2021), and social isolation (Zahedi et al., 2022) have detrimentally impacted emotional wellness across many campuses worldwide (Li .et al, 2025). At Worcester Polytechnic Institute (WPI) specifically, this issue became especially visible during the 2021 and 2022 academic years, where students surveyed reported significantly increased levels of stress, depression, and loneliness (WPI Well Being Task Force, 2022). These concerns were intensified by the social disruption caused by COVID-19 and the increased dependence on online interaction and social media.

While WPI has implemented initiatives such as wellness/ mental health days, modifications to the grading system (NR system), counseling services, and community spaces after this mental health crisis peak in 2021-22 many students continue to be plagued by feelings of isolation and disconnection.

Artistic expression and collaborative creative environments have been wielded as effective weapons in the fight to combat negative mental health and improve mental wellness holistically (Shukla et al., 2022). Public art installations and interactive artistic spaces help engage students through a wide variety of emotional, social, and creative experiences. Furthermore the engineered artistic expression aligns very closely with the values held within the institution inspiring future students with the creations of their predecessors.

This project sought to explore the combination of mechanical and electrical engineering alongside artistic design to create a unique public art installation in an attempt to create a campus environment that helps to support student well being. The project focus was on gaps in the student wellness infrastructure evaluating how a student art space might improve student interaction.

This initial phase of the project aimed to create the foundation for a synergistic and immersive wellness environment while establishing a proof-of-concept prototype that future works can expand on. Through background research, conceptual design, and mechanical creation the project aimed to create a space that invites collaboration, emotional expression, and community engagement while heralding the student mental wellness initiative on the WPI campus.

2. Background

2.1 Student Mental Health in Higher Education

Mental health concerns among college students have been on the rise over the past decade (Li et al, 2025). Students commonly experience stress related to financial pressure, academics, career uncertainty, and social isolation or expectations. Increased dependence on online communication for social interaction has led to reduced intrapersonal connectedness and sense of community (Liu et al, 2024).

Research has shown that students report symptoms of anxiety, depression, emotional burnout, and loneliness. The COVID-19 pandemic exacerbated these challenges by disrupting traditional social interaction and campus engagement (Zarowski et al., 2024) leading to heightened stress and lowered connection among students.

2.2 Mental Health at WPI

Mental health concerns at WPI became particularly visible during the 2021 and 2022 academic years. Students increasingly reported stress, emotional instability, mental fatigue, and exhaustion associated with academic pressure and social isolation. The loss of multiple students during this period marked the severity of this crisis within the campus community.

To respond, WPI implemented several mental wellness initiatives to support students including:

- Mental health days (Wellness Days)
- The modified NR system
- Expanded therapy and counseling resources
- Implemented community-centered spaces
- Promote peer support initiatives

While these programs provide necessary important services, many are mainly focused on the response to a crisis rather than proactively helping students with community and social engagement. Furthermore, these steps have somewhat improved the situation however issues continue to arise in our community.

2.3 Art and Mental Wellness

Artistic engagement has long been associated with emotional wellness, stress reduction, and self expression (Godley et al., 2025). Public immersive art spaces and creative environments promote necessary social stimuli and emotional connection within a shared sensory experience rather than simply observing passively.

Research surrounding participatory art projects demonstrates that collaborative engagement can improve connectedness, social confidence and overall wellness. Public immersive installations could be a potentially effective method because of their capacity to combine visual, auditory, and tactile stimuli to create emotionally resonant experiences that inspire reflection and connection.

These immersive installations that combine sound, light, and movement can transform abstract emotional concepts into tangible sensory experiences.

2.4 Engineering, Technology, and Society

This project examines the relationship between engineering artistic expression, technology, and emotional wellness on a university campus. Engineering as a science is traditionally associated with technical problem solving; however, this does not exclude it from shaping communities, emotional experiences, and social interactions. In fact it's quite the opposite, when implemented correctly engineering can communicate directly and indirectly with almost any group of people. Public installations demonstrate how technical systems can be used not only as a functional tool but for major emotional and social enhancement. By combining motion sensors, ambient soundscapes, lighting systems and mechanical interaction an immersive art display can create responsive environments that evolve alongside the participant; where the whole is greater than the sum of its parts.

The interference patterns installation was designed to visually and physically react between fragmented and cohesive states as a response to the participant. The shifting environmental states symbolized isolation and connection while encouraging participants to influence their space in a collaborative manner. In doing so the installation aims to reinforce the idea that emotional wellness and social connectedness are collective experiences rather than purely individual challenges.

The integration of student-built engineering systems into a public artistic setting reflects WPI’s interdisciplinary culture by merging technical creativity with a more human-oriented design. The installation very aptly defines the potential relationship between art and engineering and how it can be used to strengthen that sense of community while combating social isolation and creating an emotionally supportive campus.

2.5 Precedental Art Research

2.5.1 VisualEars: Virtual Immersive Art and Mood Enhancement

One influential precursor for this project was the *VisualEars* virtual immersive art exhibition study conducted during the COVID-19 pandemic (Hadavi et al., 2022). The aforementioned project studied whether immersive artistic experiences with blended visual art and music could produce an improvement in mood regulation and wellbeing within the general population. The exhibition utilized interconnected virtual gallery environments featuring

artworks paired with a curated musical composition to produce a multisensory emotionally connective setting. The participants were asked to complete Positive and Negative Affect Surveys (PANAS) before and after attending the exhibit where the results indicated an increase in positive affect and decreased negative affect following their attendance.

The study is a fantastic initial framework that demonstrates that immersive artistic spaces support emotional engagement, reduce stress, and improve mood in participants. Their project emphasized the importance of multisensory interaction, artistic collaboration, and environmental immersion in shaping one's mental response. These findings directly informed the ideation of the Resonant Space project by fortifying interactive and sensory-driven artistic experiences as supplementary wellness-oriented research goals.

While the VisualEars project focus was primarily on virtual engagement, the Resonant Space plans to expand upon their finding through the incorporation of physical mechanical interaction, collaborative participation, and engineering-centered artistic expression. The study served as an important precedent supporting the claim that immersive art and mental wellness go hand in hand.

2.5.2 MindClay: Interactive Installation and Therapeutic Creative Arts

Another influential research precedent examined during the development of the Resonant Space project was MindClay. MindClay is an interactive therapeutic arts initiative developed by Marco Pinter and associates that explored how interactive artistic environments and digitally mediated creative experiences can support emotional expression, mindfulness, stress reduction, and therapeutic recovery (Pinter, 2025). The initial research study of MindClay was a physical interactive installation evolving into a powerful broad digital support platform focused on therapeutic creative arts and emotional well-being.

The initial project emphasized the importance of tactile interaction, creative participation, and emotionally responsive artistic environments as tools for fostering reflection and psychological wellbeing. Unlike passive artistic observation, MindClay encouraged users to engage actively with the art through sensory interaction and creative capacity. The study highlighted how artistic technologies and their environments should be designed not only for engagement aesthetically but for self expression and direct connection.

The concepts outlined strongly influenced the ideation direction of the Resonant Space project. Similar to MindClay, this project investigates how interactive environments may encourage emotional engagement through physical participation and sensory immersion. However, Resonant Space further expands their concept towards the WPI campus environment through integration of mechanical engineering in the art alongside the collaborative public installation. The MindClay project reinforced the broader concept that immersive and interactive artistic environments may serve as accessible supplementary wellness tools capable of supporting preventative approaches to mental health and emotional well being in the community.

3. Methodology

3.1 Purpose:

The purpose of this project was to design and develop an immersive interactive art installation that combines engineering, artistic expression, and collaborative participation to support student mental wellness and social connectedness at Worcester Polytechnic Institute.

3.2 Objectives list:

The following objectives guided the development of the project by identifying distinct challenges in the student wellness at WPI. Exploring potential solutions of artistic and collaborative environments and their capacity to improve the mental health of the students on campus.

3.2.1 Objective 1: Investigate Student Health Challenges at WPI

This objective focused on understanding the major facets that contribute to burnout and decline of mental health within the WPI community. By examining student perceptions of the current wellness ventures.

Objective 1: Research Questions:

- What factors most significantly contribute to student stress and burnout?
- How do students perceive existing wellness steps?

Methods:

Background research was conducted using mental health literature, university wellness reports, and discussions regarding student psychological health. The study developed by (Watson, 1988) to measure mental health as a quantifiable metric as a response to perceived positive and negative social interactions could be used as a benchmark to evaluate how students feel pre and post visiting the installation.

3.2.2 Objective 2: Examine the Role of Artistic and Collaborative Spaces

Objective 2: Research Questions:

- How does engagement in art influence emotional wellness?
- What types of collaborative spaces encourage social interaction?
- How can interactive installation promote dialogue and collective participation?

Methods:

Research was conducted regarding public art, art therapy, and interactive community environments. Existing interactive installations and wellness-oriented spaces were analyzed for effectiveness, applicability, and feasibility on campus life at WPI. The team additionally examined a variety of impactful studies related to participatory art projects and community-centered artistic collaboration. This research helped better the understanding of how collective design and construction processes influence emotional connectedness and social support networks.

3.2.3 Objective 3: Develop a Conceptual Mechanical Art Space

Objective 3: Research Questions:

- What installation styles would best engage WPI students?
- How could art and engineering be combined to integrate well into a collaborative wellness space?
- How can sensory interaction, movement, and environmental responsiveness encourage participation and reflection.

Methods:

Conceptual designs and brainstorming sessions were used to develop installation ideas that emphasized interaction, creativity, and technical engagement. The installation design sessions incorporated mechanical systems, projections, motion sensors, ambient soundscapes, and interactive light intended to create a responsive environment connected to and with the participant. The design process additionally emphasized student participation and co-creation to reinforce the collaborative goals of the project. The final installation incorporated concepts for mechanically responsive wall components capable of extending and retracting in response to participant proximity and movement throughout the space. These dynamic physical interactions were intended to create a stronger sense of immersion and environmental responsiveness within the installation.

3.3 Design Process:

The current *Interference Patterns: A Resonant Space* project combined collaborative artistic ideation, participatory design methods, engineering problem solving, and physical fabrication to create the immersive wellness focused installation for the WPI campus. The project methods underlined the necessity of student involvement and interdisciplinary teamwork when properly portraying the campus communities interests, experiences, and emotional needs through its installation. Rather than function as a traditional one size linear engineering design

process, the project evolved rapidly through cycles of brainstorming, discussion, prototyping, construction, and refinement.

The overall design methodology emphasized modularity and space for future works. Although the current final implementation of the Resonant Space was limited primarily to the interactive wall structure, the established project provided a framework for future expansion in the Qualifying Projects space. Participatory sessions and routine communication meetings collectively shaped the development of the installation.

3.3.1 Participatory Sessions

The participatory brainstorming sessions served as the foundational starting point for the development of *Interference Patterns: A Resonant Space.* The sessions were designed to encourage joint idea proliferation; whilst providing students the space to openly discuss emotional wellness, social isolation, creativity, sensory interaction, and campus connections in an informal low-pressure environment.

The students emphasized a “students for students” philosophy in which any participant could actively contribute to the concept ideation for the installation. Rather than prescribing a fixed project rubric and design in the early development, the project intentionally promoted open-ended collaboration and co-creation in order to better understand and reflect the challenges, interests, and experiences of the WPI student body.

Throughout the brainstorming session participants explored recurring ideas and topics including:

- Emotional isolation versus social connection
- Immersive sensory interaction
- Responsive environmental systems
- Collaborative participation
- Movement and physical engagement
- Motionally expressive artistic spaces

Students contributed through sketching, group discussion, environmental mapping, and conceptual ideation exercises. These activities generated early installation concepts involving projections, 3D printed arches, ambient audioscapes, motion sensor systems, tactile interaction, and modular art. Several concepts later informed the engineering and direction of the wall prototype and broader installation framework.

Figure 1: Participatory brainstorming session used to collaboratively develop conceptual themes and interaction ideas for the installation.

The participatory sessions additionally revealed the importance of interdisciplinary teamwork during the design process. Students approaching the project from an artistic perspective emphasized the immersion, aesthetics, atmosphere, and symbolism while the students from the engineering backgrounds focused on functionality, responsiveness, fabrication, and technical implementation.The combination of the two sides appeared to strengthen the project by creating a working balance of emotional engagement and engineering feasibility.

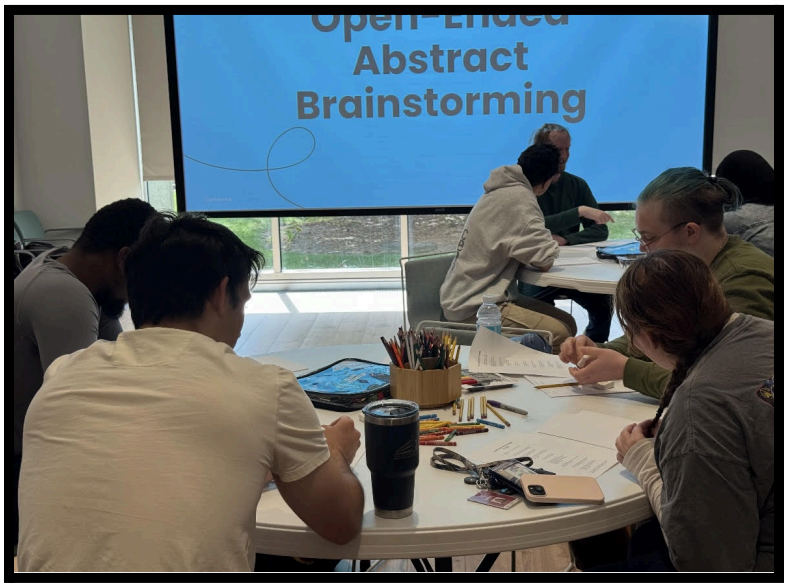


Figure 2 : Early collaborative concept ideation sessions for the Resonant Space

Figure 3: Students generate brainstorming materials exploring sensory engagement, interaction pathways, and emotional wellness themes.

The ideas generated during these participatory sessions informed the subsequent engineering development and fabrication process becoming central guides through the entire project.

3.3.2 Routine Communication Sessions

The weekly meeting sessions were conducted throughout the duration of the project in order to maintain coordination and monitor project progression while addressing challenges faced. These meetings played a critical role in keeping continuity between the conceptual, technical, and fabrication aspects of the installation.

The communication sessions involved discussions surrounding:

- Installation goals and conceptual direction
- Fabrication timelines
- Engineering feasibility
- Material selection
- Structural design
- Sensory interaction concepts
- Future expansion planning

Because the project implemented a variety of disciplines frequent communication was essential for maintaining alignment between the various components of the installation. The nature of this project required ongoing refinement as conceptual brainstorming transitioned into practical engineering concentrations.

The routine communication meetings served as check points for identifying project limitations and next steps in adapting the scope of implementation. As fabrication timelines, material constraints, and technical integration challenges became more apparent, the team used these meetings to prioritize achievable objectives while maintaining the broader long-term vision of the installation. This communication contributed to the decision to focus on the physical implementation primarily to establish the basis for future works.

Figure 4: Students presenting the project timelines during a routine meeting.

Finally, the communication sessions reinforced the collaborative nature of the project by providing continual feedback and problem solving between students. Students contributed perspectives from artistic, engineering, and experiential viewpoints, allowing the installation to grow through ongoing collaborative refinement rather than isolated design decisions.

3.4 Construction Process:

Due to the ongoing nature of the project and timeline associated with the current IQP term, this iteration primarily focused on intellectual development and construction of the foundational arches wall structures. Additional features remain areas for continued research and project iteration.

3.4.1 3D Prints

Figure 5: 3D printed wall scissor connector components fabricated for the preliminary interactive wall archetype. These components were designed to support modular assembly of this interactive installation.

The initial 3D printed connector components were designed to support the expandable scissor wall mechanism while attached to an interactive face plate. The parts were printed and intended to allow for simple and removable assembly during the iteration stages of this project. An additional printed housing sheath was printed to provide a clean mechanical interface between the placed scissor installation and the wall. The connector component was designed with a clearance for the sheath to slot into for a flush fit against the locking each other in place.

Figure 6: Prototype motor sheath housing developed to encase and stabilize the actuator system within the wall structure during preliminary assembly and testing.

3.4.2 Component Assembly

Following solidworks fabrication and printing the linear actuators were combined with their housing sheaths to eliminate movement and evaluate mechanics and structural alignment within the wall prototype. The actuator assemblies served as the platform for controlled motion of the expandable scissor mechanisms.

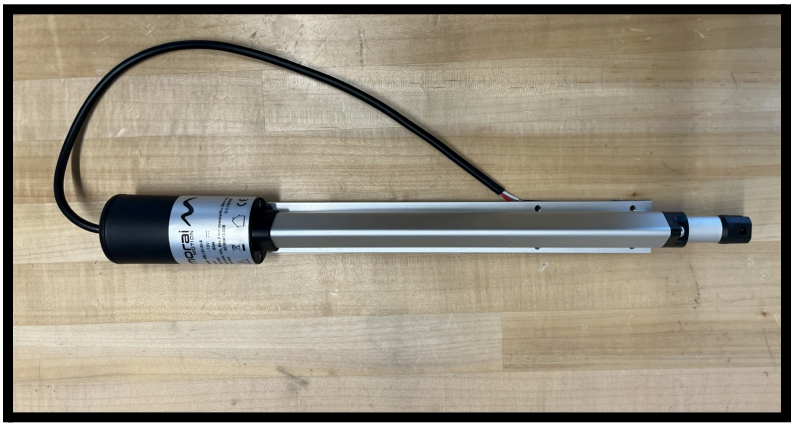


Figure 7: Linear actuator used within the prototype wall system to provide controlled mechanical movement for the expandable scissor mechanism.

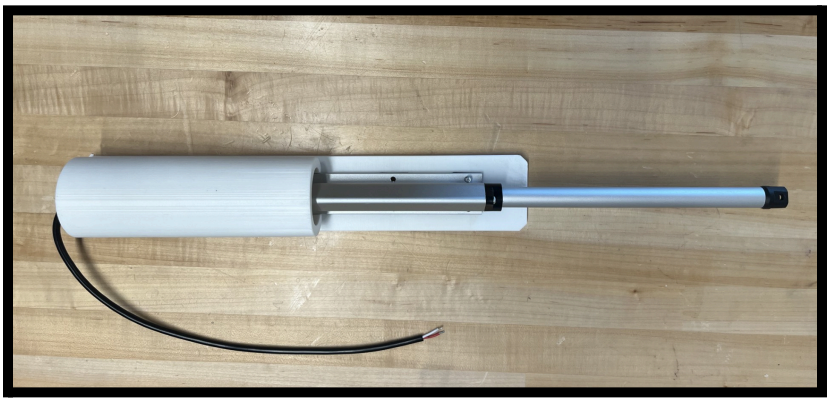


Figure 8: Preliminary actuator integration within the custom motor sheath assembly designed for structural support and modular attachment to the wall framework.

Mechanical linkage systems were then assembled to investigate expandable movement patterns and modular structural behavior. The scissor mechanisms were selected due to their ability to create large-scale motion while maintaining compact structural geometry.

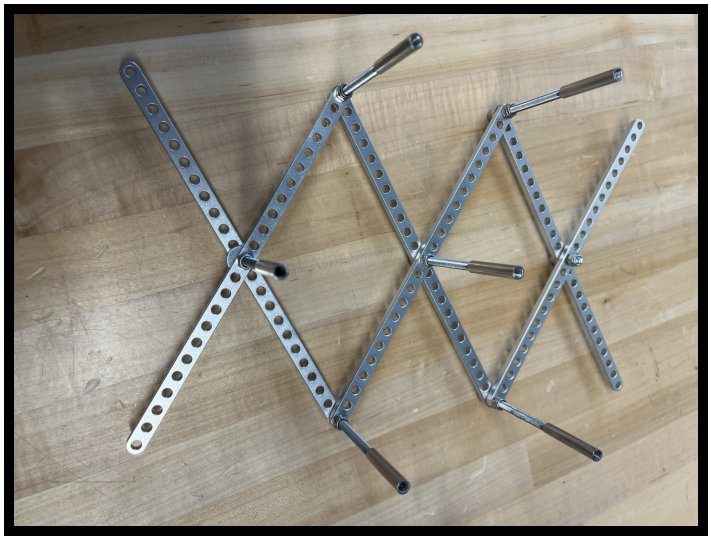


Figure 9: Scissor linkage mechanism used to create expandable movement within the interactive wall prototype. The linkage system enables controlled structural expansion and contraction during user interaction.

Figure 10: Preliminary assembly of the 3D printed wall connector integrated with the scissor linkage support structure.

Preliminary integration testing was conducted by combining the scissor linkage assemblies with the 3D printed support structures. Where this phase focused on mechanical stability and the feasibility of synchronized expansion across multiple wall units.

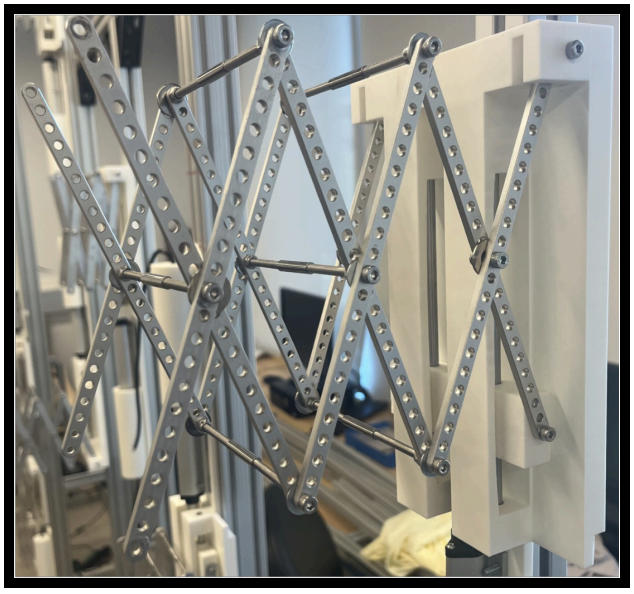


Figure 11: Scissor wall unit assembly integration of the mechanical linkage system with the 3D printed connector mounted to the wall support framework during prototype construction.

3.4.3Wall Creation

After individual component testing and assembly, the modular wall units were integrated into a larger structural prototype. The wall creation process emphasized modularity and future growth to allow later Qualifying Project teams to expand the installation with additional features.

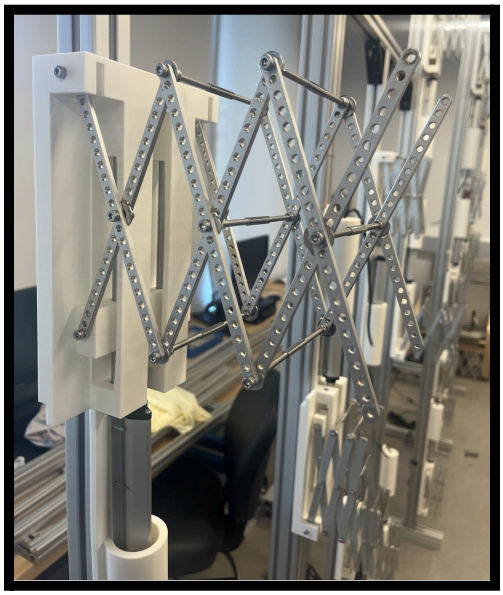


Figure 12: Early-stage construction of the interactive wall prototype illustrating integration of modular supports, actuator systems, and expandable scissor mechanisms.

Figure 13: Alternate perspective of the prototype wall assembly showing mechanical alignment and structural positioning of the expandable linkage system.

Structural supports and mounting systems were incorporated to improve stability and actuator alignment throughout the larger wall assembly. This phase also provided insight into future considerations involving maintenance or material replacement and long-term installation feasibility.

Figure 14: Front-facing structural support assembly developed to stabilize the modular wall prototype during preliminary testing and fabrication.

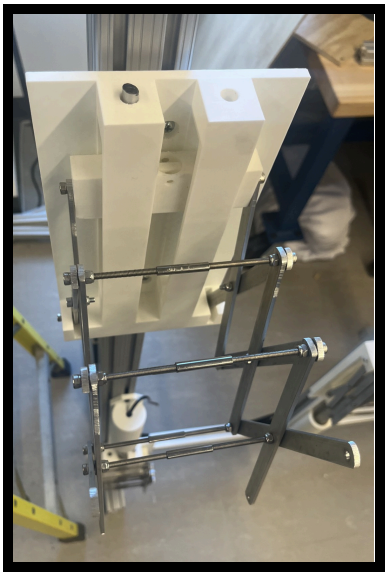


Figure 15: Top-down structural framework and mounting configuration used to support actuator integration and mechanical movement within the prototype wall system.

The completed foundational wall prototype represented the culmination of the current project iteration. While the prototype does not yet contain the full immersive and interactive functionality originally envisioned, it establishes an initial proof-of-concept framework capable of supporting future systems integration.

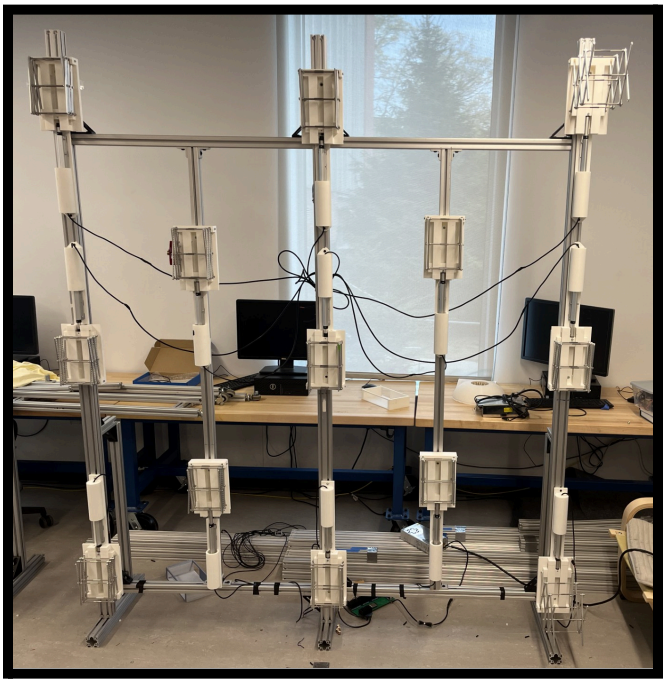


Figure 16: Completed foundational wall prototype developed during the current IQP iteration. The modular structure serves as a preliminary proof-of-concept for future immersive and interactive wellness installations at WPI.

4. Findings

Due to the scope and timeline of the project the current iteration focused on conceptual development and construction of the foundational archway wall prototype; therefore, the observations discussed in this chapter should be considered preliminary and exploratory rather than conclusive. This chapter presents the major findings developed throughout our research, conceptual development, and partial implementation of the Resonant Space. Analysis of student wellness concerns, immersive artistic environments, collaborative engineering and the construction of an interactive prototype wall was conducted. The final evaluation of the relationship between engineering, public art, social connection, and student mental wellness is still underway.

The findings suggest that immersive and collaborative art spaces may provide meaningful supplementary support to the current campus wellness initiatives at WPI. The support from the project shines through by encouraging emotional engagement, interpersonal interaction, and community participation. Additionally, the design and construction process demonstrated how engineering-centered artistic projects align with WPI's technical culture while simultaneously fostering creativity in students.

4.1 Wall Construction Demonstrated Feasibility of an Interactive Wellness Space

Although a complete installation of the fully developed space will need numerous more projects similar to this one; the successful construction of the interactive wall prototype has demonstrated the feasibility of developing a larger immersive wellness installation on the WPI campus. The wall served as an initial foundation for exploring how engineering criteria and artistic movements can be involved and evolved on the campus.

The construction process provided preliminary support for several important design concepts. The wall construct provides a working platform capable of supporting future technical and artistic elements. The modular nature of the structure additionally serves as the precursor for future expansion into the larger immersive experience.

Additionally the complete wall provides an early visual and conceptual representation of the larger installation goals. Even without the full integration of responsive systems and immersive effects in the current iteration of the project it helped demonstrate how engineering structure and artistic presentation could coexist within a collaborative wellness context.

This construction process furthers the role of engineering as a collaborative and creative discipline at WPI. Mechanical construction, prototyping, fabrication, and iterative problem solving became central components of the project's artistic expression rather than existing away from it. That collaboration of technical construction and emotional design aligned very closely with the broader interdisciplinary goals of the installation. Overall, the completed arch wall

prototype supplied a proof-of-concept basis for supporting future development of a larger immersive installation at WPI.

4.2 Continued Steps and Preliminary Observations

Although the current iteration of the Inference Patterns: A Resonant Space primarily focused on the development and fabrication of the interactive arch wall prototype, the project established several important preliminary observations regarding immersive artistic environments, collaborative design, and engineering-centered wellness ventures. The completed wall structure and participatory design session indicated future viability of constructing a larger interactive exhibit at WPI.

The positive response generated during the brainstorming sessions and meetings engaged students in a similar fashion consistent with the end goals of the project. The collaborative nature of the brainstorming process itself appeared to encourage communication, shared creativity, and interpersonal engagement in a lower-pressure environment. These observations give some merit to the project hypothesis where art through an engineering environment may help strengthen feelings of community and connectedness among students.

Additionally, the arch wall construction prototype process supports the idea that engineering systems could serve as both structural and artistic components in an art exhibition. The wall provided a stable basis for future integration of technical effects structures for participant interaction. The modularity of the materials and structure of the arch established the feasibility of iterative expansion through future Qualifying Projects and interdisciplinary continuation projects at WPI. This process also revealed several important considerations for future implementation. The integration of the mentioned systems requires significant background or coordination between mechanical engineering, electrical engineering, coding, and artistic design. Future efforts towards continuation will therefore benefit from increased collaboration across the disciplines and longer term fabrication timelines.

The continued steps were identified as critical in the advancement of the installation towards a final fully immersive interactive exhibit. Future efforts should focus on:

- Integrating projection and lighting systems into the existing wall structure
- Implementing motion-responsive interaction and environmental feedback systems
- Developing immersive ambient soundscapes
- Conducting user interaction and engagement studies
- Expanding tactile and kinetic interaction elements
- Evaluating long-term emotional wellness outcomes associated with installation use

Additionally, future Qualifying Projects could investigate how students interact with the immersive environment over an extended period of time and whether repeated engagement added to measurable improvements in emotional wellness, stress reduction, or feelings of social

connectedness. Both qualitative and quantitative feedback from participants would provide stronger evidence regarding the effectiveness of engineering-centered art as a preventative wellness initiative.

The current project therefore serves as the proof-of-concept and the foundational framework for future continuation efforts. While the full installation was not completed in the scope of this project conceptual development, collaborative participation, and arch construction strongly support continued exploration of immersive engineering-centered artistic environments as potential tools in the promotion of student wellness and strengthening campus community.

5. Recommendations & Deliverables

This chapter lists the recommendations from preliminary research, conceptual development, and partial implementation conducted throughout the WPI administration to improve mental wellness amongst its student body. The included recommendations are divided into immediate next phases and the short and long term recommendations in order to provide both directly achievable steps and broader university goals for future planning directions.

5.1 Next Steps in the Resonant Space Development

Although the current project successfully established the foundation and prototype wall structure for *Interference Patterns: A Resonant Space*, several phases of development are necessary in order to realize the immersive wellness environment. The following next steps outline the recommended continuation pathway for the future Qualifying Projects to expand the installation. These future phases are intended to progressively transition the installation from proof-of-concept into a fully interactive exhibit that supports emotional engagement and community interaction in the long-term.

5.1.1 Code Base Sensor Testing

The following IQP continuation team should prioritize the development and testing of the artworks foundational sensor and interaction systems. While the current project established the framework for responsive environmental interaction, limited project length prevented full integration and calibration of these systems in the arch walls.

Future development should focus on creating an expandable coding infrastructure that is capable of connecting and supporting:

- Motion detection systems
- Proximity sensors
- Projection responsiveness

- Interactive lighting systems
- Ambient audio triggering
- Participant-based environmental feedback

The next team should explore how different interaction thresholds and environmental responses influence the participants. This variation in motion sensitivity, lighting transition, and audio response may affect the immersion and usability of the installation. The successful implementation of these foundational systems will serve as a critical step toward transforming the installation from a static structural prototype into a fully immersive responsive environment.

5.1.2 Sensor Integration and Build Finalization

Following successful sensor testing, the next project group should focus on the responsive systems integration directly into the walls prototype; and finalize the physical structure into the connected archway. This phase should emphasize the implementation of motion-responsive projections, dynamic lighting, ambient audio systems, and additional kinetic interaction components.

The team should additionally evaluate sensor placement, power distribution, accessibility, durability, and participant safety while keeping the modularity of the installation for future expansion and mobility. The goal of the structure is to transition the project from preliminary into the full functional experience it was designed to be.

5.1.3 Interaction Survey

Following implementation of responsive sensory systems, participant interaction studies should be conducted to denote and store behavioral responses to the installation. A created Positive and Negative Affect Survey (PANAS) as outlined in (Hadavi et al., 2022) should be used before and after participant interaction to measure potential changes in emotional state and overall engagement.

Further survey questions should evaluate the following:

- Perceived immersion
- Social connectedness
- Accessibility
- Stress reduction
- Environmental comfort

Observational analysis may be used to study participant interaction and collaborative engagement with the space for quantifying research goals.

5.1.4 Sensory Additions and Surveyed Reactions

Future continuation teams should expand the installation through additional sensory systems such as:

- Synchronized ambient soundscapes
- Reactive lighting environments
- Tactile interaction surfaces
- Vibration feedback systems
- Environmental projections
- Dynamic color transitions
- Responsive kinetic structures

Where each sensory element should be evaluated both independently and collectively in order to determine how these combinations affect:

- Emotional immersion
- Stress levels
- Participant comfort
- Interpersonal interaction
- Environmental engagement

The survey feedback should shape the combinations that are implemented and take into account comfort and accessibility of the participants while eliminating any concerns that may arise.

5.2 Short Term Recommendations

The following short term recommendations focus on the initial steps that WPI would be able to implement to begin developing an interactive mechanical art and wellness space on campus. These recommendations emphasize community feedback and student collaboration to evaluate feasibility and improve student engagement while establishing a basis for further expansion.

5.2.1 Develop a Small-Scale Pilot Installation

WPI should begin by implementing a temporary or small-scale interactive mechanical art installation in a highly trafficked campus location such as the Campus Center, Innovation Studio, library entrance or on quad space near or on the seal. Starting with a pilot installation would allow the university to evaluate the exhibits reception before committing significant financial or infrastructural resources to a permanent installation.

The pilot installation should prioritize accessibility and student turnout. Where potential features could include but not be limited to:

- Mechanically interactive moving components to compliment the created archway
- Tactile kinetic constructs
- Light responsive systems
- Student controlled moving display similar to the archway
- Visual or audio art display booths

This pilot approach would allow WPI to gather the feedback regarding preferred aesthetics, interaction styles, and ideal locations while identifying safety and maintenance measures early in development.

5.2.2 Integrate Student Feedback and Wellness Input

Before permanent implementation, WPI should conduct additional surveys, interviews, and focus groups with members of the university community to get a better grasp on what types of spaces could be beneficial.

Special attentions to factors including but are not limited to:

- Sensory and tactile considerations
- Accessibility and inclusivity
- Stress-reducing design and environment
- Desired interaction levels
- Preferred campus locale
- Concerns of overstimulation and crowding

Counseling and wellness professionals could be included or interviewed to provide optimal stress mitigation. These individuals can also provide feedback to ensure the installation aligns with broader student wellness initiatives whilst avoiding unintentional stressful or exclusionary environments.

5.3 Long-Term Recommendations For The Wellness Space

The following is a list of long-term recommendations that focus on the continued development and institutional integration of interactive artistic initiatives at WPI. They aim to support lasting cultural change by promoting further research on the incorporation of art-engineering-centered community spaces into the university.

5.3.1 Permanent Interactive Wellness Space

Following the anticipated success of the short-term goals of the pilot implementation WPI should consider establishing a permanent interactive mechanical art display space. WPI administration after analysis of student and faculty feedback could implement a wellness space designed to immerse the students with interactive art for students by students.

Unlike the traditional passive art displays like those at Worcester Museum of Art this permanent installation should be designed as an evolving environment based on community collaboration encouraging:

- Direct student participation
- Curated social interaction
- Creative expression
- Technical curiosity
- Emotional decompression

The space could include rotating installations and opportunities for students to regularly contribute new artwork or mechanical systems. Long-term adaptability would help maintain continued student interest while allowing the space to evolve alongside campus culture.

5.3.2 Expansion Research on Engineering Connected to Art and Mental Wellness

This project denotes the potential relationship between engineering-centered artistic engagement and mental stability on the WPI campus. However additional research still needs to be conducted to evaluate the longer term effectiveness of this standard of project. Future studies should investigate:

- Long-term student engagement trends
- Measurable emotional wellness metrics
- Improvements in social connectedness
- Tolerable stress levels for the student body
- Effects of immersive art installations in public spaces on campus

Additional research could compare different types of installations, interaction pathways, and environmental designs to determine which approach is most feasible in promoting student health as a whole.

5.3 Deliverables

The *Interference Patterns: A Resonant Space* project produces several deliverables intended to establish both a physical and conceptual foundation for future creation of an

immersive wellness-oriented exhibit on the WPI campus. While the full immersive environment was not completed within the project timeline, the deliverables provide a scalable basis for future Qualifying Projects and other continuation efforts.

The primary physical deliverable was the successful design and fabrication of the interactive arches wall prototype. The wall served as the foundational framework structure for the eventual larger installation concept and provides a strong feasibility of engineering integration in artistic expression for the improved mental wellness on the WPI campus. The modular nature of the wall additionally supports further future progression through connection of sensory systems, projections, soundscapes, and tactile or kinetic interactive elements.

Additional deliverables included:

- Participatory brainstorming session materials,
- Conceptual sketches and installation ideation documents
- Engineering and fabrication concepts
- Modular structural designs
- Preliminary coding frameworks for responsive interaction systems and their connected electrical systems
- Documentation regarding future continuation pathways

The project additionally produced conceptual foundations outlining future implementation phases for immersive sensory interaction systems, participant engagement studies, and emotional wellness evaluation methods (PANAS). These provide continuation teams with an organized pathway for expanding the installation toward its intended fully immersed state.Collectively, these deliverables establish a foundation for future interdisciplinary projects exploring the relationship between engineering-centered artistic environments, emotional wellness, and community engagement within university settings.

5.4 Conclusions

This precursor phase of the project suggests that interactive artistic engineering focussed environments may positively influence student wellness by promoting creativity, collaboration, and social connectedness. The research and prototype development demonstrated potential for immersive wellness spaces to complement current campus mental health ventures.

Bibliography

Barbayannis, G., Bandari, M., Zheng, X., Baquerizo, H., Pecor, K. W., & Ming, X. (2022). Academic Stress and Mental Well-Being in College Students: Correlations, Affected Groups, and COVID-19. *Frontiers in psychology*, *13*, 886344. https://doi.org/10.3389/fpsyg.2022.886344

Chen, L., & Zeng, S. (2021). The Relationship Between Intolerance of Uncertainty and Employment Anxiety of Graduates During COVID-19: The Moderating Role of Career Planning. *Frontiers in psychology*, *12*, 694785. https://doi.org/10.3389/fpsyg.2021.694785

Hadavi, S., Kennedy, K. G., Mariotti, G., & DeSouza, J. F. X. (2022). VisualEars: How an immersive art exhibit impacts mood during the COVID-19 pandemic. *Frontiers in psychology*, *13*, 910767. https://doi.org/10.3389/fpsyg.2022.910767

Godley, L., O'Hayer, C. V., Davis, R., Wakschal, E., Nurse, C., Ross, W., Spraker, A., Godshall, N., & Frasso, R. (2025). "University stress" exploring the potential impact of an immersive art experience on a college campus. *Frontiers in psychology*, *16*, 1592649. https://doi.org/10.3389/fpsyg.2025.1592649

Li, Q., Li, J., & Fan, Y. (2025). Addressing mental health in university students: a call for action. *Frontiers in public health*, *13*, 1614999. https://doi.org/10.3389/fpubh.2025.1614999

Liu, Q., & Lin, D. (2024). The impact of distance education on the socialization of college students in the Covid-19 era: problems in communication and impact on mental health. *BMC medical education*, *24*(1), 575. https://doi.org/10.1186/s12909-024-05551-7

Moore, A., Nguyen, A., Rivas, S., Bany-Mohammed, A., Majeika, J., & Martinez, L. (2021). A qualitative examination of the impacts of financial stress on college students' well-being: Insights from a large, private institution. *SAGE open medicine*, *9*, 20503121211018122. https://doi.org/10.1177/20503121211018122

Pinter M., Brown C., Ho P. (2025). *MindClay*: From Interactive Installations to a Digital Platform for Therapeutic Creative Arts. *Leonardo* 2025; 58 (6): 574–581. doi: https://doi.org/10.1162/leon_a_02714

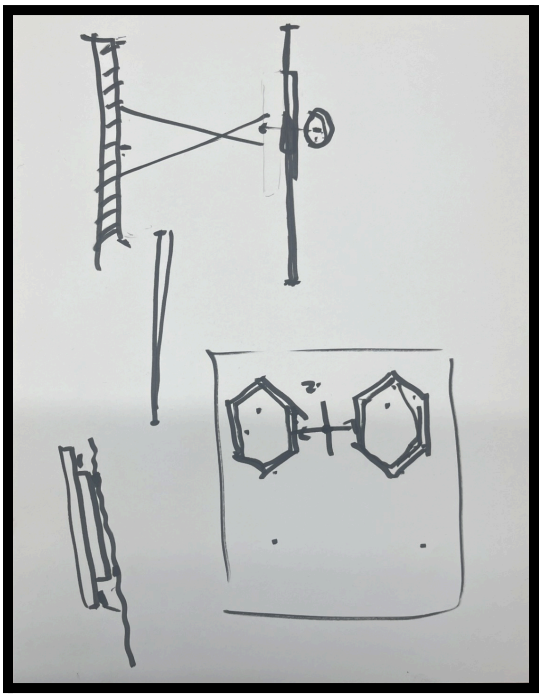
Shukla, A., Choudhari, S. G., Gaidhane, A. M., & Quazi Syed, Z. (2022). Role of Art Therapy in the Promotion of Mental Health: A Critical Review. *Cureus*, *14*(8), e28026. https://doi.org/10.7759/cureus.28026

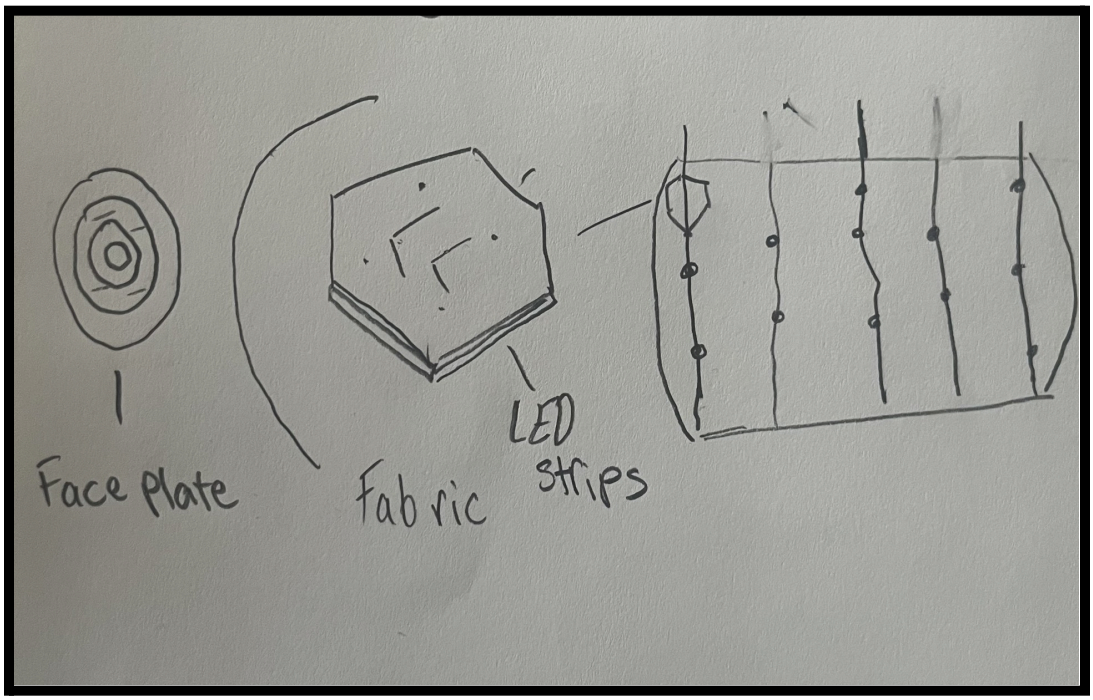
Watson D. (1988). Intraindividual and interindividual analyses of positive and negative affect: their relation to health complaints, perceived stress, and daily activities. *Journal of personality and social psychology*, *54*(6), 1020–1030. https://doi.org/10.1037//0022-3514.54.6.1020

Zahedi, H., Sahebihagh, M. H., & Sarbakhsh, P. (2022). The Magnitude of Loneliness and Associated Risk Factors among University Students: A Cross-Sectional Study. *Iranian journal of psychiatry*, *17*(4), 411–417. https://doi.org/10.18502/ijps.v17i4.10690

Zarowski, B., Giokaris, D., & Green, O. (2024). Effects of the COVID-19 Pandemic on University Students' Mental Health: A Literature Review. *Cureus*, *16*(2), e54032. https://doi.org/10.7759/cureus.54032

Appendix A: Participatory Brainstorming Sessions Images





Professor Farzin
Graduate researchers in interactive design
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Center for Wellbeing
Participatory Design Sessions
Guidelines/Ground rules

Appendix B: Archetype Interaction Plate Designs